

PRAVINKUMAR K

Full Stack Developer | React.js | Next.js | TypeScript | Node.js | MongoDB

Coimbatore, India | 8220839631 | pravinkumar671231@gmail.com

linkedin.com/in/pravinkumar31 | pravinkumar-portfolio-rho.vercel.app

EXECUTIVE SUMMARY

Full Stack Developer experienced in turning requirements into functional and reliable web applications. Comfortable working across frontend, backend, database, and deployment layers to build complete features. Focused on simple, maintainable code, practical solutions, and delivering reliable applications.

CASE STUDIES

Proof of Work

SERVER-SIDE PRODUCT SEARCH

MongoDB · Mongoose · TypeScript · REST API

Problem: Client-side product search required downloading the full product catalog to the browser, increasing data transfer and reducing scalability as the catalog grew.

Action:

- Added the `q` query parameter to the Get Products API for server-side keyword search.
- Implemented case-insensitive partial matching across name and description using MongoDB regex.
- Used MongoDB `$facet` to fetch paginated products and the total matching count in a single aggregation query.

Result: server side product search with efficient pagination and total result counting in a single database query.

DYNAMIC FILTERING

TypeScript · Express.js · Mongoose · MongoDB · REST API · MongoDB Aggregation

Problem: Static filter definitions are difficult to maintain when different product categories have varying brands, specifications, variants, and price ranges.

Action:

- Built a `GET /api/categories/:slug/filters` API to dynamically generate category-specific filter options.
- Derived brands, price ranges, specification ranges, and variant axes from active category products using backend aggregation logic.
- Used Mongoose + MongoDB with parallel queries to efficiently build the complete filter configuration.

Result: Generate dynamic, backend-driven category filters, eliminating hardcoded frontend configurations.

SECURE R2 UPLOAD WORKFLOW

TypeScript · Express.js · Cloudflare R2 · Presigned URLs · REST API · RBAC

Problem: Product and category images need secure file uploads without exposing storage credentials or unnecessarily routing large files through the application server.

Action:

- Implemented admin-only presigned upload APIs for Cloudflare R2 with MIME type validation and UUID-based object keys.
- Added upload tracking and single-use object-key validation to ensure only server-issued upload keys can be attached to entities.
- Implemented image count validation and primary-image normalization to enforce consistent image data across products, brands, and categories.

Result: Secure R2 uploads with direct storage transfer, unique object keys, and consistent image validation.

DYNAMIC CATEGORY SPECIFICATION SYSTEM

Mongoose · MongoDB · TypeScript · REST API

Problem: Product categories require flexible specifications with different field types, units, options, required fields, and validation rules without hardcoding category-specific schemas.

Action:

- Built a category-level specification API at PATCH `/api/admin/categories/:id/specifications` supporting grouped fields, typed values, enums, units, required and filterable attributes.
- Implemented Mongoose schemas and service-layer validation for duplicate groups/fields, type validation, required fields, and enum options.
- Added reference-aware update/delete guards to prevent removing specification fields or groups that are already used by products.

Result: Flexible category-specific product specs with consistent validation and safe schema changes.

PROJECTS & EXPERIENCE

TechCart - Full-Stack E-Commerce Platform

Built a full-stack e-commerce platform with separate Buyer App and Admin Console, comprising 26 frontend routes and 83 REST API endpoints.

- **Buyer App:** Built responsive product discovery, category browsing, product details, filtering, sorting, pagination, and other customer-facing workflows using Next.js, React.js, TypeScript, and Tailwind CSS.
- **Admin Console:** Developed reusable interfaces for product, category, brand, specification, variant, and image management with responsive forms and workflows.
- **API & State Management:** Integrated REST APIs using Redux Toolkit and RTK Query, handling authenticated requests, queries/mutations, caching, loading/error states, and server-state synchronization across both applications.
- **Dynamic UI:** Implemented frontend filtering and sorting using backend-generated brands, price ranges, specifications, and variants.
- **Responsive Design:** Built reusable, mobile-responsive UI components and layouts using Tailwind CSS across Buyer and Admin applications.

GitHub: github.com/Pravin671231/TechCart | Live Demo: tech-cart-buyer-app.vercel.app

MovieNest - Movie Discovery Application

- **Responsive UI:** Built a mobile-first, responsive movie discovery interface using React.js, TypeScript, and Tailwind CSS across mobile, tablet, and desktop layouts.
- **Custom API Hooks:** Developed reusable React Hooks for asynchronous TMDb API requests, handling loading, error, and data states with clean state management

GitHub: github.com/Pravin671231/movies | Live Demo: im-db-movie.netlify.app

Web Development Intern — Techvolt Software Company, Coimbatore

- Developed responsive web pages using HTML5 and CSS3 following modern frontend and responsive design practices.
- Built clean, structured layouts, strengthening skills in UI implementation and dev workflows.

TECHNICAL SKILLS

Frontend	TypeScript, JavaScript (ES6+), HTML5, CSS3, React.js, Next.js, Tailwind CSS, Redux Toolkit, RTK Query, Context API
Backend & APIs	Node.js, Express.js, REST APIs, Zod, MongoDB, Mongoose, MongoDB Atlas
Tools & Version Control	Git, GitHub, Postman, Vercel, Render
Familiar	Vitest, React Testing Library, MSW, Supertest, Docker, Docker Compose, GitHub Actions, CI/CD, Cloudflare R2

EDUCATION & CERTIFICATIONS

- B.Sc. Botany — Bharathiyaar University, Coimbatore | 2021
- Certified MERN Stack Developer — IDM TechPark, Coimbatore | February 2025
- AI-Augmented Development: Build, Test & Deploy Apps — LearnzConnect | June 2026

LANGUAGES

Tamil, English